Cloud Computing
Unit 3: Building Cloud Networks

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Evolution from Managed service providers (MSP) to Cloud Computing

- Managed Service Provider
  - Third-party contractor that delivers network-based services, applications and equipment to enterprises, residences or other service providers
  - Can be hosting companies or access providers that offer IT services such as fully outsourced network management arrangements, including IP telephony, messaging and call center management, virtual private networks (VPNs), managed firewalls and monitoring/reporting of network servers
  - Can offer services such as alerts, security, patch management, data backup and recovery for different client devices: desktops, notebooks, servers, storage systems, networks and applications
  - Act as an extension of your IT department, taking care of routine IT infrastructure monitoring and management around the clock—freeing up your IT staff to focus on higher-value projects

Evolution from Managed service providers (MSP) to Cloud Computing

- After years of investing in infrastructure and business model changes to deliver remote managed services, MSPs are seeing the cloud as a threat to their livelihood
- In many cases, the cloud requires no remote infrastructure or on-premise equipment
- The fear is the cloud could render MSPs obsolete
- Cloud computing could also be a managed service. Cloud computing consists of security monitoring, storage management, network administration and the all managed services need not necessarily be cloud computing
- Cloud computing puts its efforts in creating a technical solution. It is a technical model that delivers technical access to the computing resources
- Managed services, on the other hand, is a contract based relationship. The definition and delivery of the service is done on a repeating revenue basis. It means that managed services are recurring revenues from well-defined services that are predictable.

Evolution from Managed service providers (MSP) to Cloud Computing

- There are a lot of similarities that exist between a managed service and cloud computing
- While cloud computing is generally a technical implementation that decides how the infrastructure and applications are to be delivered over the private and public networks, it can also be a business model
- The same contract agreement holds ground in a cloud computing relationship. Managed services can also be a technical implementation
- Managed service providers (MSPs) offer one of the oldest forms of cloud computing
- Basically, a managed service is an application that is accessible to an organization's IT infrastructure rather than to end users
- For the cloud, the biggest evolutionary jump began with managed service providers (MSPs) and their motivation to satisfy and retain customers paying monthly recurring fees.

Single-Purpose Architectures to Multipurpose Architectures

- In the early days of MSPs, the providers would actually go onto customer sites and perform their services on customer-owned premises
- Over time, these MSPs specialized in implementation of infrastructure and quickly figured out ways to build out data centers and sell those capabilities off in small chunks commonly known as monthly recurring services, in addition to the basic fees charged for ping, power, and pipe (PPP)
- Ping refers to the ability to have a live Internet connection, power is obvious enough, and pipe refers to the amount of data throughput that a customer is willing to pay for

Single-Purpose Architectures to Multipurpose Architectures

- The PPP part of the charge was built into the provider's monthly service fee in addition to their service offerings
- Common services provided by MSPs include remote network, desktop and security monitoring, incident response, patch management, and remote data backup, as well as technical support
- An advantage for customers using an MSP is that by purchasing a defined set of services, MSPs bill a flat or near-fixed monthly fee, which benefits customers by having a predictable IT cost to budget for over time
- Step forward to today and we find that many MSPs now provide their services remotely over the Internet rather than having to sell data center space and services or perform on-site client visits (which is time-consuming and expensive)

Data Center Virtualization

- Data center virtualization is a method of moving information storage from physical servers to virtual ones, often in a different location
- In the past, large companies would keep physical servers on site that held huge amounts of corporate information
- These servers were expensive, both to purchase and maintain
- With data center virtualization, it became possible to separate both the hardware and location from the data
- This cuts costs and increases the data's availability
- Data center virtualization actually comes from a combination of two different technologies; high-speed data transfer and server virtualization. Without both of these components, data center virtualization becomes highly impractical

Data Center Virtualization

- Benefits of Data Center Virtualization
  - Enables the consolidation of physical servers, slashing the costs of operating a data center. This includes reducing the costs of server upgrades, management, power, space, and storage
  - Reduction in data center space and in data center equipment such as PDUs, air conditioning units, etc
  - Reduction in the number of network, HBAs and SAN switches
  - Provides true high-availability for all servers without requiring duplicate hardware and clustering software
  - Facilitates true disaster recovery for all servers
  - Eliminates the need for maintenance windows for physical server troubleshooting or upgrades and enables faster server provisioning
  - Enhances security and provides regulatory compliance benefits

Data Center Virtualization

- The cloud evolved from the roots of managed service provider environments and data centers and is a critical element of next-generation data centers when compared to the MSPs they evolved from
- Today, customers no longer care where the data is physically stored or where servers are physically located, as they will only use and pay for them when they need them

Cloud Data Center

- A cloud data center has three distinct characteristics that differentiate it from traditional DC
- It is sold on demand, typically by the minute or the hour
- It is elastic -- a user can have as much or as little of a service as they want at any given time
- The service is fully managed by the provider (the consumer needs nothing but a personal computer and Internet access)
- The multiplier effect of Internet data, trends of enterprises information transformation and tremendous load bring forth traditional DC huge challenges: how to reduce operation and maintenance cost, how to meet demand of high capacity, high security and high efficiency

Cloud Data Center

- Attributing to resource on-demand, flexible and dynamic structure, cloud computing is the right technology to resolve all these issues
- By continuously improving core technologies including virtualization, elastic computing and high-density computing etc, cloud vendors have creatively developed Modular cloud computing Data Center of trusty, efficient, smart, ultra-bandwidth and green end to end solution
- Cloud Data Center is now evolving beyond being merely a model of technology delivery to becoming a new operating model where business decision makers are empowered to procure infrastructure on demand, and where IT becomes an internal service provider delivering increased business agility without compromising security or control

Traditional Corporate Data Center
Vs
Cloud Data Center

Traditional Corporate Data Center	Cloud Data Center
Thousands of different applications	Few applications (maybe even just one)
Mixed hardware environment	Homogeneous hardware environment
Multiple management tools	Standardized management tools
Frequent application patching and updating	Minimal application patching and updating
Complex workloads	Simple workloads
Multiple software architectures	Single standard software architecture

Service Oriented Architectures (SOA)

- Service-Oriented Architecture (SOA)
  - Is an architectural style
  - Applications built using an SOA style deliver functionality as services that can be used or reused when building applications or integrating within the enterprise or trading partners.
  - Uses open standards to integrate software assets as services
  - Standardizes interactions of services
  - Services become building blocks that form business flows
  - Services can be reused by other applications

SOA Defined

- "SOA is the architectural style that supports loosely coupled services to enable business flexibility in an interoperable, technology agnostic manner. SOA consists of a composite set of business-aligned services that support a flexible and dynamically re-configurable end-to-end business process realization using interface-based service descriptions."

From a paper by Borges, Holley and Arsanjani.

What is a Service?

- A service is a reusable component that can be used as a building block to form larger, more complex business-application functionality.
- A service may be as simple as “get me some person data,” or as complex as “process a disbursement.”
- A service provides a discrete business function that operates on data.
- Its job is to ensure that the business functionality is applied consistently, returns predictable results, and operates within the quality of service required.

Characteristics of a Service

- Supports open standards for integration: Although proprietary integration mechanisms may be offered by the SOA infrastructure, SOA's should be based on open standards. Open standards ensure the broadest integration compatibility opportunities
- Loose coupling: The consumer of the service is required to provide only the stated data on the interface definition, and to expect only the specified results on the interface definition. The service is capable of handling all processing (including exception processing).

Characteristics of a Service

- Stateless: The service does not maintain state between invocations. It takes the parameters provided, performs the defined function, and returns the expected result. If a transaction is involved, the transaction is committed and the data is saved to the database.
- Location agnostic: Users of the service do not need to worry about the implementation details for accessing the service. The SOA infrastructure will provide standardized access mechanisms with service-level agreements.

SOA

- One of the key benefits of a service oriented approach is that software is designed to reflect best practices and business processes instead of making the business operate according to the rigid structure of a technical environment.
- A service-oriented architecture is essentially a collection of services
- A service is, in essence, a function that is well defined, self-contained, and does not depend on the context or state of other services.
- The services generally communicate using standard protocols, which allows for broad interoperability
- New services can be added or created without affecting existing services

SOA

- The first service-oriented architectures are usually considered to be the Distributed Component Object Model (DCOM) or Object Request Brokers (ORBs), which were based on the Common Object Requesting Broker Architecture (CORBA) specification
- The introduction of SOA provides a platform for technology and business units to meet business requirements of the modern enterprise
- With SOA, your organization can use existing application systems to a greater extent and may respond faster to change requests

SOA

- Differentiating between SOA and cloud computing can be confusing because they overlap in some areas but are fundamentally different
- SOA delivers web services from applications to other programs, whereas the cloud is about delivering software services to end users and running code

Combining Cloud and SOA

- Cloud services benefit the business by taking the best practices and business process focus of SOA to the next level
- These benefits apply to both cloud service providers and cloud service users.
- Cloud service providers need to architect solutions by using a service-oriented approach to deliver services with the expected levels of elasticity and scalability
- Companies that architect and govern business processes with reusable service-oriented components can more easily identify which components can be successfully moved to public and private clouds

SOA Planning

- Planning for Capacity
  - set up an initial infrastructure and establish a baseline of capacity
- Planning for Availability
  - includes performing a business impact analysis (BIA) and developing and implementing a written availability plan
- Planning for SOA Security
  - The foundations include public key infrastructure (PKI), the common security authentication method Kerberos, XML (Extensible Markup Language) encryption, and XML digital signatures
  - Three main areas of concern:
    - Message-Level Security
    - Security-as-a-Service
    - Declarative and Policy-Based Security

Open Source Software in Data Centers

- The term *open source* broadly describes a general type of software license that makes source code available to the public without significant copyright restrictions
- Recently, more than just a few users have become strong advocates that open source can and does work in the data center environment.
- An enterprise service bus (ESB) refers to a software architecture construct that is typically implemented by technologies seen as a type of middleware infrastructure.
- ESBs are usually based on recognized standards and provide fundamental services for complex architectures via an event-driven and standards-based messaging engine (called the bus since it transforms and transports the messages across the architecture).

Some Points to Know about Open Source Software

- The author or holder of the license of the source code cannot collect royalties on the distribution of the program.
- The distributed program must make the source code accessible to the user.
- The author must allow modifications and derivations of the work under the program's original name.
- No person, group or field of endeavor can be denied access to the program.
- The rights attached to the program must not depend on the program's being part of a particular software distribution.
- The licensed software cannot place restrictions on other software that is distributed with it

Advantages of OSS

- Open-source software is free to use, distribute, and modify. It has lower costs, and in most cases this is only a fraction of the cost of their proprietary counterparts
- Open-source software is more secured as the code is accessible to everyone. Anyone can fix bugs as they are found, and users do not have to wait for the next release. The fact that is continuously analyzed by a large community produces secure and stable code.
- Open source is not dependent on the company or author that originally created it. Even if the company fails, the code continues to exist and be developed by its users. Also, it uses open standards accessible to everyone; thus, it does not have the problem of incompatible formats that exist in proprietary software
- Lastly, the companies using open-source software do not have to think about complex licensing models and do not need anti-piracy measures like product activation or serial number.

Disadvantages of OSS

- The main disadvantage of open-source software is not being straightforward to use. Open-source operating systems like Linux cannot be learned in a day. They require effort and possibly training from your side before you are able to master them. You may need to hire a trained person to make things easier, but this will incur additional costs
- There is a shortage of applications that run both on open source and proprietary software; therefore, switching to an open-source platform involves a compatibility analysis of all the other software used that run on proprietary platforms. In addition, there are many ongoing parallel developments on open source software. This creates confusion on what functionalities are present in which versions.
- Lastly, many of the latest hardware are incompatible to the open-source platform; so you have to rely on third-party drivers.